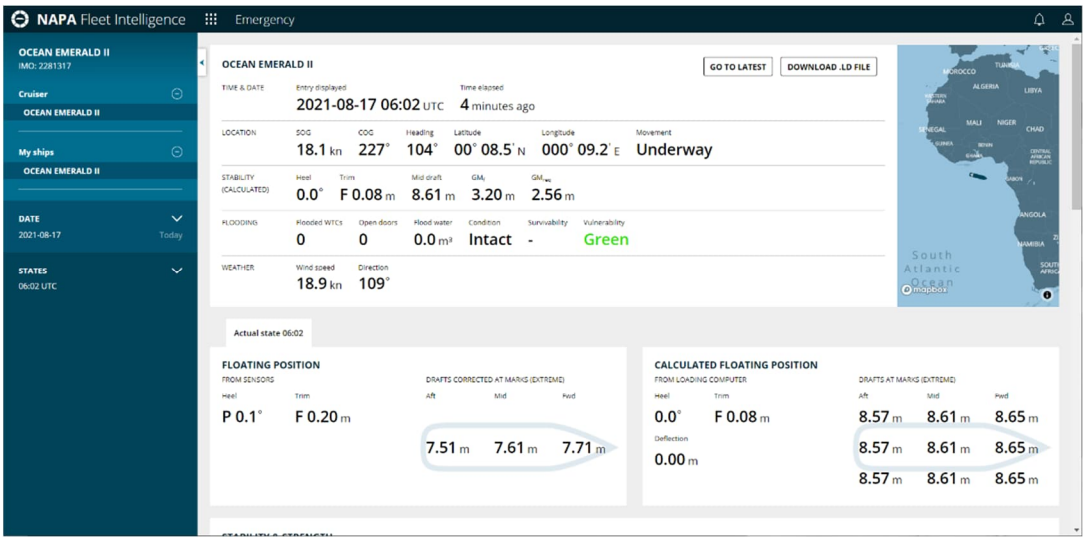
NAPA Fleet Intelligence

Safety Solutions Modules



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NAPA FLEET INTELLIGENCE – FLEET, STABILITY AND LOGBOOK DATA SHARING THROUGH CLOUD SERVICES

1 INTRODUCTION

NAPA Fleet Intelligence is NAPA's cloud solution enabling to combine and share data from the fleet. It provides different modules enabling for shore-based users to monitor accurate real-time information on the ship status produced by the following onboard NAPA solutions: NAPA Stability, NAPA Loading Computer, NAPA Emergency Computer and NAPA Logbook. The modules continuously receive data from NAPA system onboard the ship. Third party weather and AIS data is also included.

The access to ship's real time status and to earlier recorded statuses is a valuable asset also in normal everyday operation as well as exceptional situations. The information can be made available anywhere to e.g. fleet super-intendents, fleet operations centers or third parties such as emergency response services. The data can be accessed from a web browser, and thus doesn't require any additional shoreside software installations.

Data can be shared with all relevant internal and external stakeholders via the cloud platform in an easy and secure way. These stakeholders may include company's own safety teams and DPAs, external Emergency Response Services, salvage companies or the authorities. The access to ship's real time status and to earlier recorded statuses is a valuable asset also in normal everyday operation.

NAPA Fleet Intelligence for Safety Solutions Modules can be completed with additional modules such as ship voyage performance monitoring and analysis.

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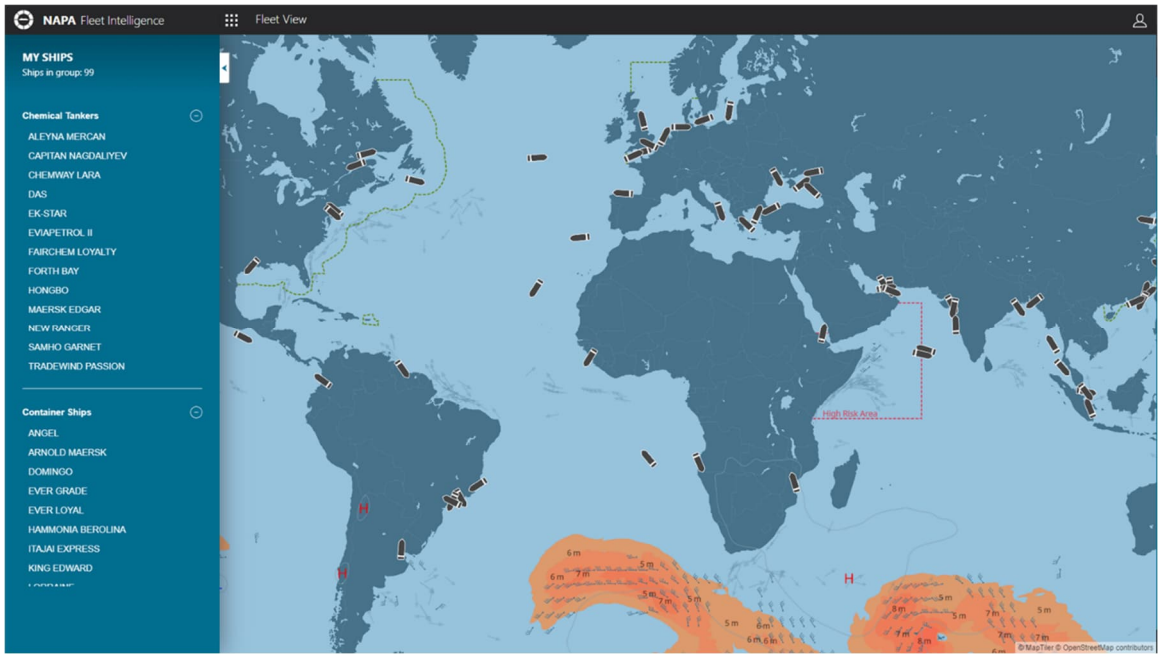
2 NAPA FLEET INTELLIGENCE SAFETY SOLUTIONS MODULES

2.1 Fleet View Module

The Fleet View module provides the latest available location of the fleet as well as the AIS and weather information. The map also presents Emission Control Areas and High Risk Areas. A single ship status can be viewed by clicking on the ship name or icon for more information on the latest status (route, speed, weather). By zooming in to the map, further details on the weather can be viewed.



NAPA Fleet Intelligence Fleet View Module

2.2 Stability Monitor Module

The NAPA Fleet Intelligence Stability Module provides an overall view of the fleet stability status and alerts. The main view will have all ships listed categorizing them to two lists depending on if the ship in in a normal condition or if attention is required. Different tags will be added to the list in case there's a warning regarding the stability or flooding situation of the ship. In the main view the fleet list can be filtered by showing only ships with a selected criteria, such as warnings, ships moving or depending on data availability.

A drop down menu on each ship will present more numeric and graphical information on the ship. This information includes:

- stability criteria (GM, KG, GM limit, Bending and Shear Force)
- safety criteria from NAPA Emergency Computer (Vulnerability and survivability level, flood water amount and status of water tight doors)
- floating position (calculated and sensor drafts, center of gravity and seawater density)
- deck information (tanks and compartments)
- location (movement, SOG, COG, lattitude, longitude)

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- mass loads (displacement, lightweight, deadweight, mass loads with unknown deadweight, further details on unknown deadweight, list of mass loads)
- tanks data (total of fuel, ballast, fresh water and gray water; 48 hour tank history)
- supplemetal information on the ships (ship background information, stability booklet information)

Details on ship stability data

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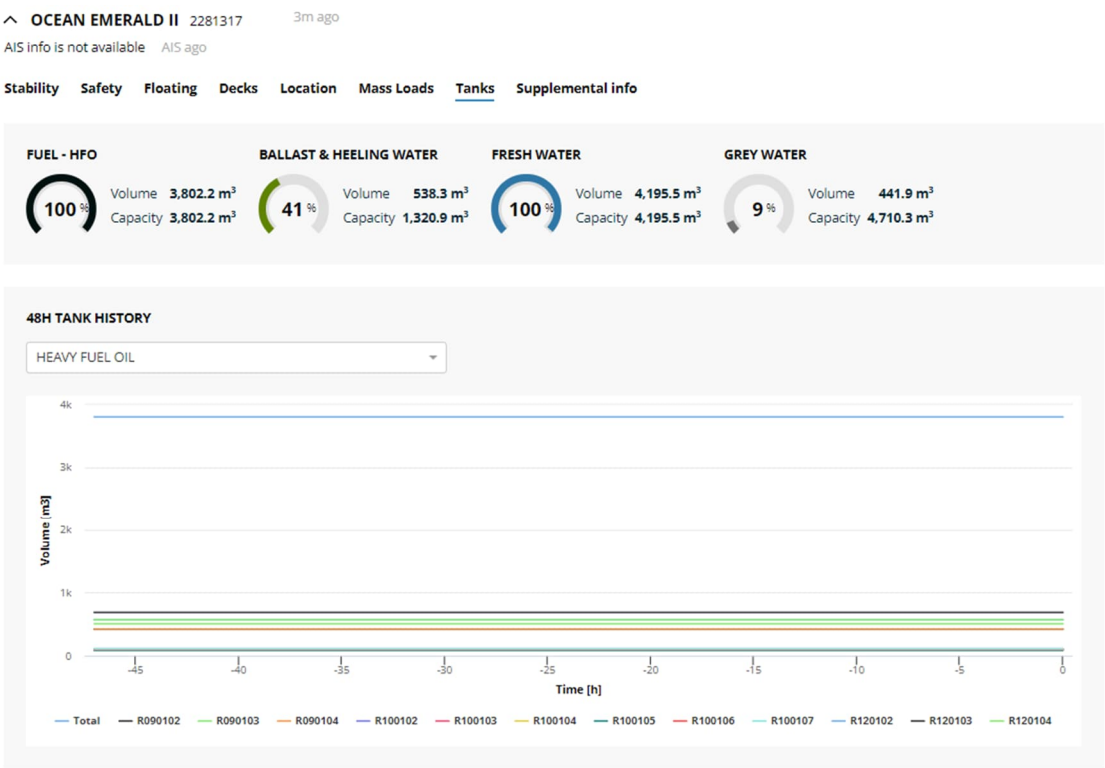

Details on ship tanks data

2.3 Emergency Module

This Emergency Module includes details for the selected ship's position, loading condition, stability and strength status. In case of flooding, the module displays the flooding state and progress. If the ship has the NAPA Emergency Computer system, also the flooding prediction for next 3 hours is shown. This data is of fundamental help for decision making during emergencies. The automatic data transfer will reduce need for manual communications between ship and shore. This reduces workload and minimizing possibilities for mistakes in the communications.

Within the user interface the detailed ship's status can be viewed for any moment of time. The corresponding LD file can be downloaded for in-depth analysis in NAPA Loading Computer or NAPA Emergency Response.

For non-emergency use, the system will display and collect, in high frequency, the operational history data of the ship. This data can be utilized for various purposes and analytics.

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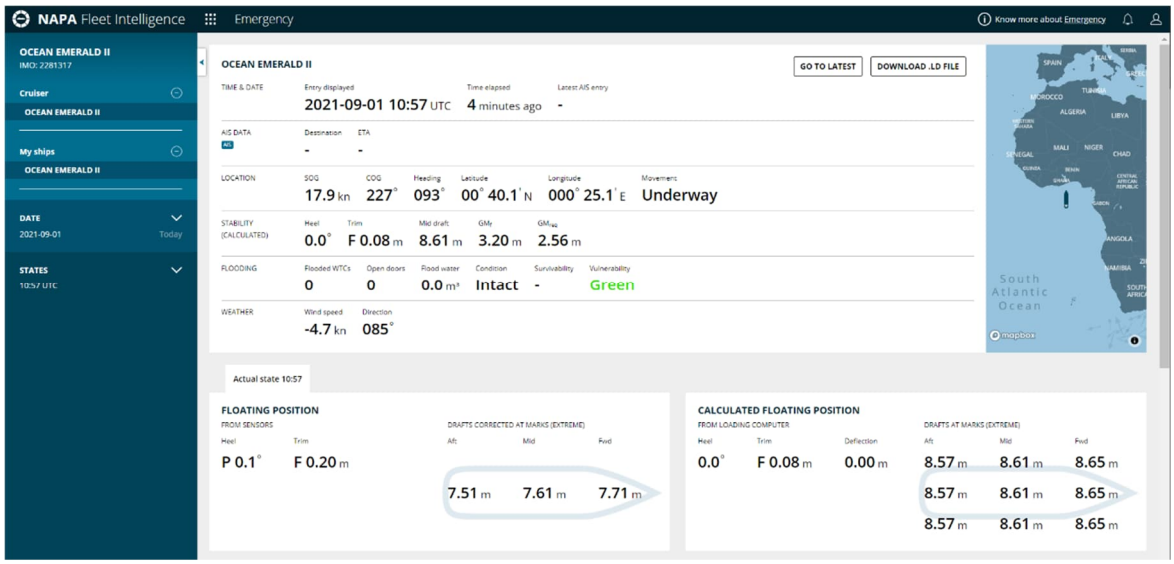
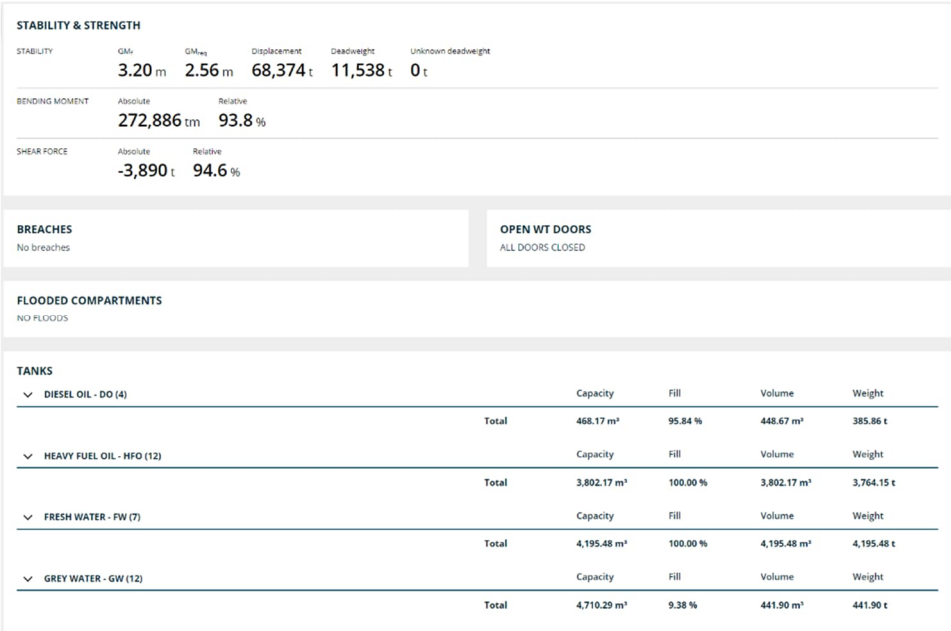


NAPA Fleet Intelligence Emergency Module View



STABILITY & STRENGTH

STABILITY	GM	GM	Displacement	Deadweight	Unknown deadweight
	3.20 m	2.56 m	68,374 t	11,538 t	0 t

BENDING MOMENT	Absolute	Relative
	272,886 tm	93.8 %

SHEAR FORCE	Absolute	Relative
	-3,890 t	94.6 %

BREACHES
No breaches

OPEN WT DOORS
ALL DOORS CLOSED

FLOODED COMPARTMENTS
NO FLOODS

TANKS

		Capacity	Fill	Volume	Weight
DIESEL OIL - DO (4)					
	Total	468.17 m³	95.84 %	448.67 m³	385.86 t
HEAVY FUEL OIL - HFO (12)					
	Total	3,802.17 m³	100.00 %	3,802.17 m³	3,764.15 t
FRESH WATER - FW (7)					
	Total	4,195.48 m³	100.00 %	4,195.48 m³	4,195.48 t
GREY WATER - GW (12)					
	Total	4,710.29 m³	9.38 %	441.90 m³	441.90 t

Details on ship stability data, flooding status and tank information

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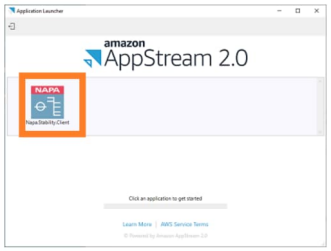
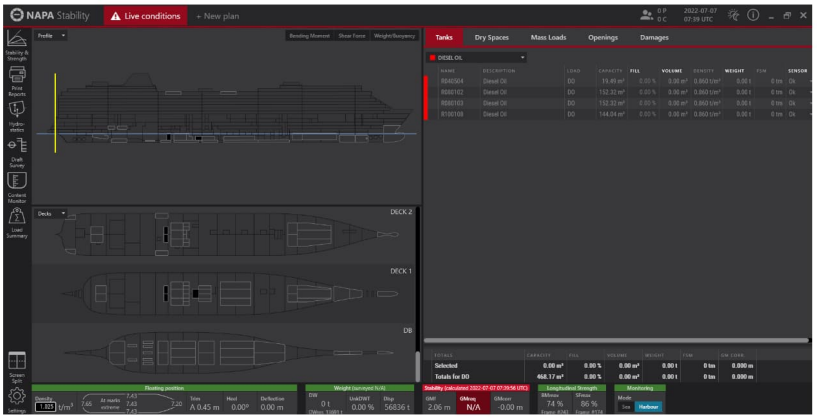
2.4 Stability Computer Module

NAPA Stability is available as a module for cloud based use. The use includes the same main functions as for NAPA Stability onboard without interfering with onboard processes. An Amazon AppStream client download is required as this will stream the server application to your desktop. One setup on a workstation will let access all the fleet ships. The client presents you the user interface of the server application. In order to enable control of the application, mouse and keyboard control events are sent from the streaming client window to the server. No content will be transferred from your workstation to the server.





2.5 Logbook Module

NAPA Logbook is one of the industry leader in ship electronic logbooks. Onboard it can replace any paper logbook to an electronic version. All the NAPA Logbook data can be immediately transferred to the cloud and made availabel for shoreside personel.

The logbook module enables to view, filter and find all data recorded in NAPA Logbook onboard the selected ships. The following filtering criteria are available:

- ship
- date range
- books
- entries
- field names
- free text search
- unapproved entries
- deleted entries
- entries with attachements

The entries can be vizualized with just the entries main data (ship, time, logbook, entry name, officer who logged the entry and last modification time) or full details of the entry.

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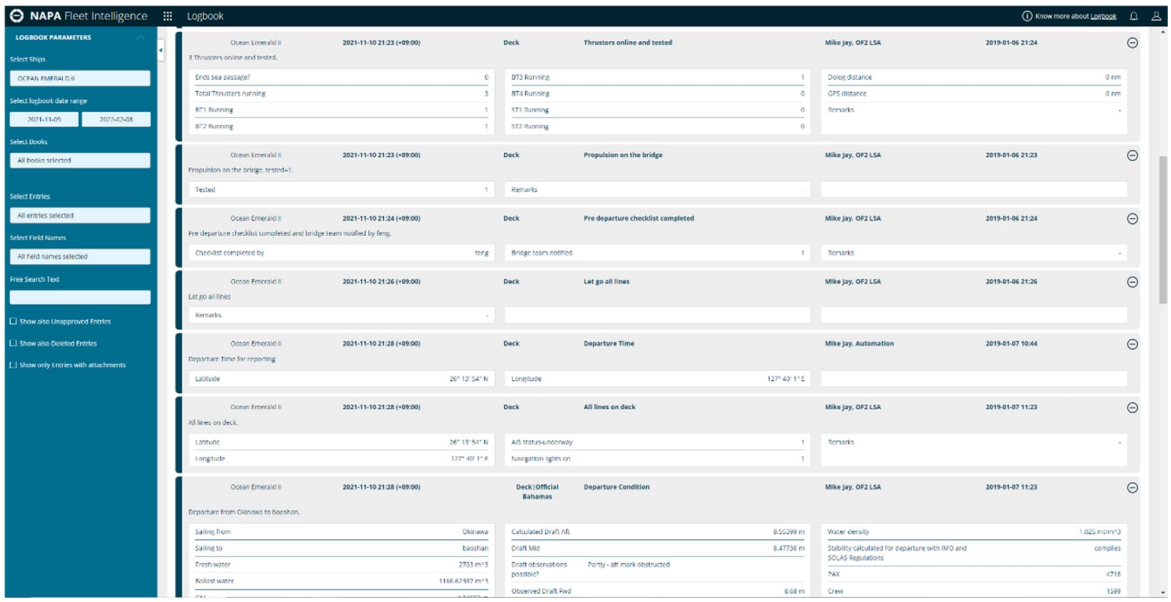
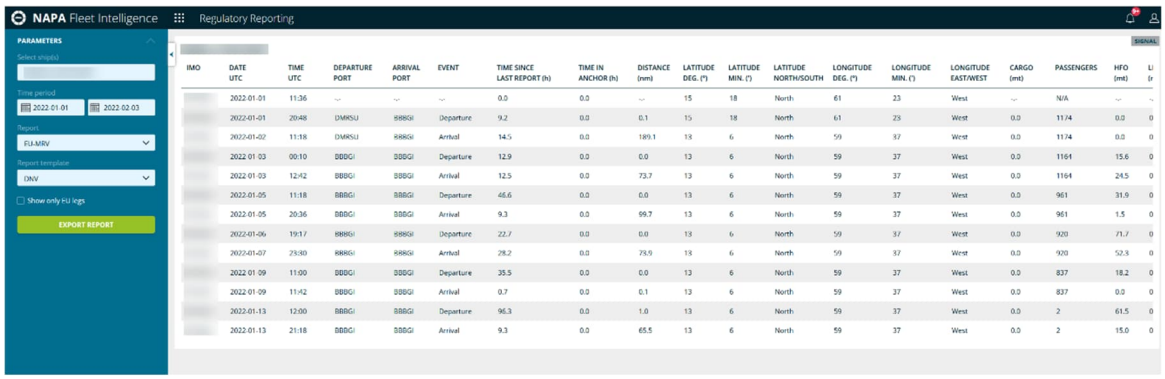


2.6 Regulatory Reporting Module

The Regulatory Reporting Module helps achieving compliancy with EU-MRV (European Union Monitoring Reporting and Verification) and IMO-DCS (International Maritime Organisation Data Collection System) reporting.

The module allows the selection of the ship, the time period and the report type (EU-MRV or IMO-DCS). The data can be filtered to show only EU legs. The module also ofers the possibility to export the data to a DNV report template format (CSV/EXCEL).



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3 USER INTERFACE

The web-based user interface is accessible in NAPA Fleet Intelligence platform, https://fleetintelligence.napa.fi/. NAPA Fleet Intelligence is a web-based platform for vessel performance monitoring, reporting, and analysis, combining various data sources with highly accurate ship performance models to create insights and advice for improved performance.

The cloud platform can utilize data from other sources and combine it with the data from ship to form a comprehensive situational image of the ship. The additional data are e.g. maps, weather data, other vessels around the ship.

4 DATA SOURCES

Onboard data collection:

NAPA Stability **NAPA Loading Computer**	Loading condition (tanks and mass loads, stability and strength)
	Draft, trim, heel sensor data
	Flooding sensor data
	Watertight door statuses
NAPA Emergency Computer (if available)	Damage/breach locations and size estimates
	Flooding prediction results for next 3 hours
NAPA Logbook	Logbook entries from the different selected logbooks
NAPA Voyage Reporting	Data for MRV / DCS reporting
Navigation interfaces (if available)	Ship navigational data (lat/long, speed, course, heading, wind speed & direction)

Other data sources:

AIS	Global database of ship AIS data
Weather & current	Weather, sea current and tide (nowcast and forecasts)

5

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6 IT REQUIREMENTS FOR THE ONBOARD MODULE

6.1 Host computer

NAPA will install the onboard module to the existing NAPA server onboard. There is no need for new computers. The module is a lightweight plug-and-play module and does not require software updates on any other NAPA modules onboard. Installation can be done remotely if a remote connection is provided.

- Web browser: Chrome, Firefox, Edge supported (latest versions recommended). Please note! Internet Explorer is not supported
- Internet connection: Broadband internet recommended
- Computer: Modern computer, Screen size 13 inch or larger, screen resolution 1600 x 900 or larger, RAM: Min 4 GB

6.2 Internet access requirements

The onboard module will need access to Internet for sending the data to the cloud system. The onboard module is sending data in 5 minute intervals, thus the Internet access needs to be continuously open.

Parameter	Requirement
Protocol	https outgoing (encrypted TLS1.2)
Target address	safety-fleet-intelligence.napaoffice.com
Port	443

6.3 Data communication size estimates

The amount of data sent to cloud is small. Estimated data amount is shown below:

Mode	Typical max. communication data
Normal operation mode	1 MB / day and NAPA Logbook data
Emergency operation mode	1 MB / hour (Only if ship has NAPA Emergency Computer and ship is flooded.)

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7 DATA SECURITY

The data communications from ship to shore and the cloud system is developed following industry best practises regarding information security.

The communication in internet and data in storage in cloud is encrypted.

The cloud platform is hosted in industry leading cloud environment (AWS) and the software and security configuration of the NAPA solution is regularly audited by independent 3rd party information security specialists.

NAPA's quality management system is ISO 9001 certified and NAPA also has laso been granted an ISO 27001 certification for information security management.

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